

CHALLENGE – 15 – CRYPTOGRAPHY

credstuff

Medium

Cryptography

picoCTF 2022

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Hints ?

Description

1

We found a leak of a blackmarket website's login credentials. Can you find the password of the user **cultiris** and successfully decrypt it?

Download the leak [here](#).

The first user in **usernames.txt** corresponds to the first password in **passwords.txt**. The second user corresponds to the second password, and so on.

20,218 users solved

77%

Liked

picoCTF{FLAG}

Submit Flag

Solution Walkthrough

1. Analyze the Provided Data

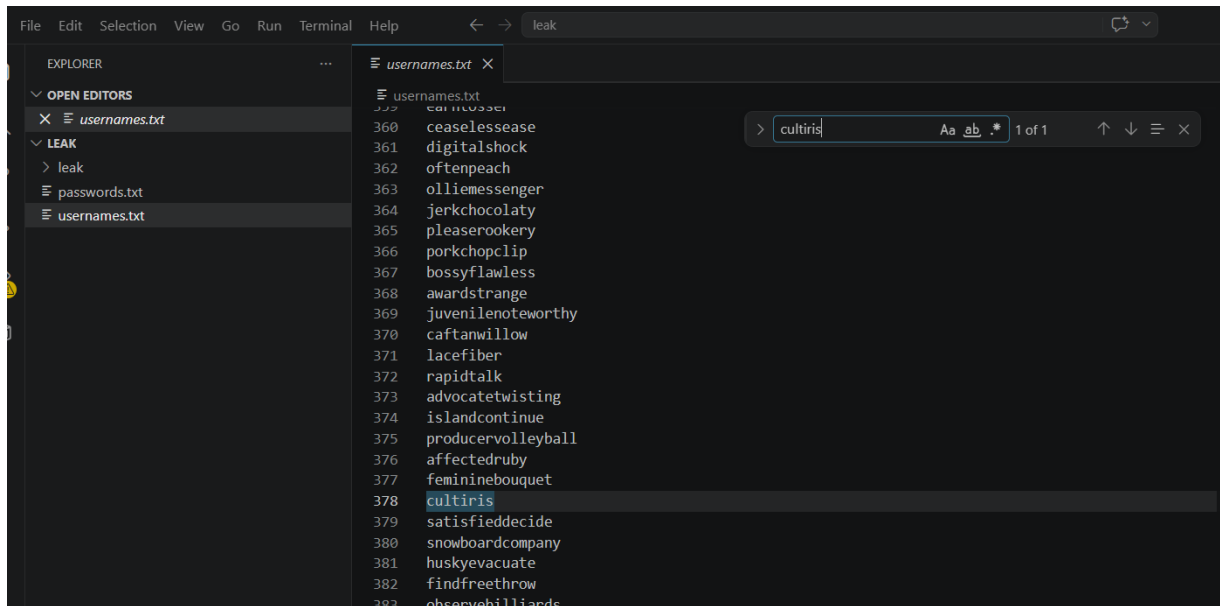
The challenge provides a "leak" consisting of two files: **usernames.txt** and **passwords.txt**. The description states that the first username corresponds to the first password, the second to the second, and so on.

2. Locate the Target User

The target user is **cultiris**.

- By searching **usernames.txt**, the user **cultiris** is found on **line 378**.

- Following the logic provided, the corresponding password is also located on **line 378** of passwords.txt.

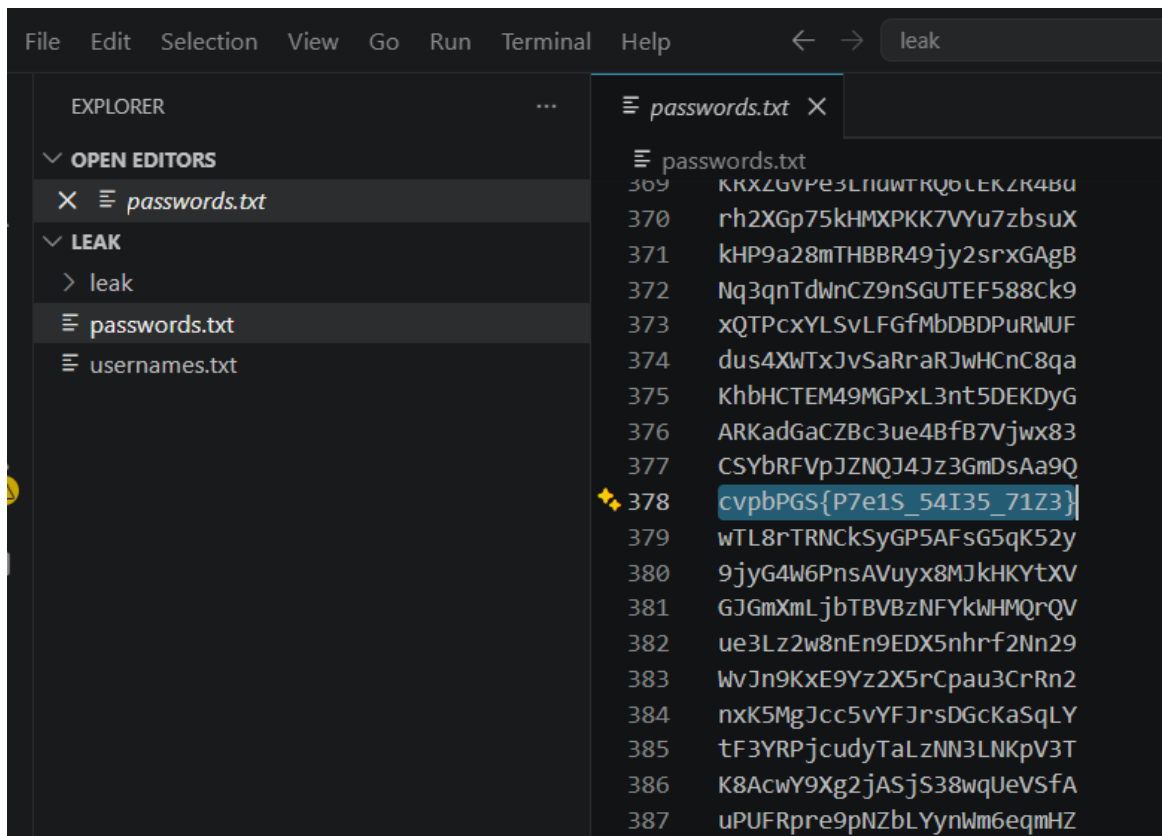


The screenshot shows a code editor with the 'usernames.txt' file open. The Explorer panel on the left shows the file structure. The main editor area displays the contents of 'usernames.txt', with line 378 highlighted. The text on line 378 is 'cultiris'. A search bar at the top right shows the search term 'cultiris'.

3. Extract the Ciphertext

From line 378 of the password file, the encrypted flag is identified as:

cvpbPGS{P7e1S_54135_71Z3}



The screenshot shows a code editor with the 'passwords.txt' file open. The Explorer panel on the left shows the file structure. The main editor area displays the contents of 'passwords.txt', with line 378 highlighted. The text on line 378 is 'cvpbPGS{P7e1S_54135_71Z3}'. A search bar at the top right shows the search term 'leak'.

4. Decryption via ROT13

The format of the ciphertext (a string of letters followed by brackets) strongly suggests a substitution cipher, specifically **ROT13**. ROT13 is a simple Caesar cipher that replaces a letter with the 13th letter after it in the alphabet.

To confirm this, the ciphertext was processed using **CyberChef's** "ROT13 Brute Force" tool.

- **Input:** cvpbPGS{P7e1S_54135_71Z3}
- **Operation:** ROT13 (Shift of 13)

gchq.github.io/CyberChef/#recipe=ROT13_Brute_Force(true,true,false,100,0,true,"")&input=Y3ZwYlBHU3tQN2UxU181NEkz...

Last build: 2 days ago

Recipe

ROT13 Brute Force

☒ Rotate lower case chars ☒ Rotate upper case chars

☐ Rotate numbers Sample length 100 Sample offset 0

☒ Print amount Crib (known plaintext string)

Input

cvpbPGS{P7e1S_54135_71Z3}

Output

Amount = 12: ohbnBSE{B7q1E_54U35_71L3}

Amount = 13: picoCTF{C7r1F_54V35_71M3}

Amount = 14: qjdpDUG{D7s1G_54W35_71N3}

Amount = 15: rkeqEVH{E7t1H_54X35_71O3}

Amount = 16: slfrFWI{F7u1I_54Y35_71P3}

Amount = 17: tmgsGXJ{G7v1J_54Z35_71Q3}

Amount = 18: unhtHYK{H7w1K_54A35_71R3}

Amount = 19: voiuIZL{I7x1L_54B35_71S3}

Amount = 20: wpjvJAM{J7y1M_54C35_71T3}

Amount = 21: xqkwKBN{K7z1N_54D35_71U3}

Amount = 22: yr1xLCO{L7a1O_54E35_71V3}

Amount = 23: zsmymDP{M7b1P_54F35_71W3}

Amount = 24: atnzNEQ{N7c1Q_54G35_71X3}

Amount = 25: buoaOFR{O7d1R_54H35_71Y3}

STEP BAKE! ☒ Auto Bake

5. Final Result

The brute-force output at **Amount 13** reveals the flag: **picoCTF{C7r1F_54V35_71M3}**

Reflection

The "credstuff" challenge highlights a common real-world scenario: handling leaked credential databases. It emphasizes the importance of data organization—knowing how to index and match related datasets (usernames to passwords).

From a cryptographic standpoint, this challenge reinforces the recognizability of the **ROT13/Caesar cipher**. Even when the internal contents are obfuscated with numbers and underscores, the distinctive "picoCTF" prefix (which becomes "cvpbPGS" when shifted by 13) serves as a clear indicator of the encryption method used. This "known plaintext" signature is often the first thing a security analyst looks for when encountering unfamiliar ciphertext.